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Name James Stuart

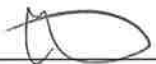
Midterm Examination 2
CHE 230
Molecular Systems in Chemical Engineering

UIC Chemical Engineering Department

04/27/2026

Time: 70 minutes

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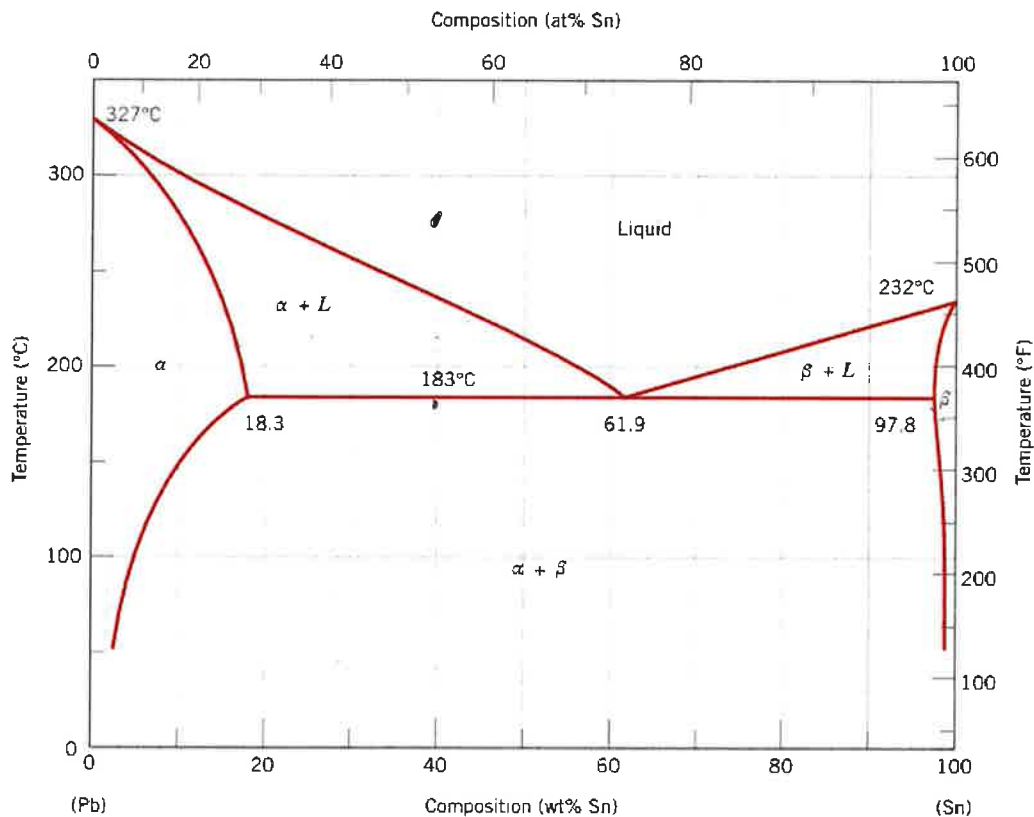
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Date 4-27-26

Part A: Phases

1. A Lead-Tin (Pb-Sn) alloy with an overall composition of 40 wt% Sn is slowly cooled from the liquid state to just below the eutectic temperature of 183 °C. At this temperature, the maximum solid solubility of Sn in the α phase is 18.3 wt% Sn, the maximum solid solubility of Pb in the β phase is 97.8 wt% Sn, and the eutectic composition is 61.9 wt% Sn.

- Determine the mass fractions of the primary α microconstituent and the eutectic microconstituent. (6 points)
- Determine the mass fraction of the *total* α phase in the alloy. (4 points)



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2. Derive the lever rule (8 points)

$$X_\alpha = \frac{X_{SnB} - C_0}{X_{SnB} - X_{SnA}}$$

$$X_{SnA} + X_{SnB} = C_0$$

$$X_\alpha + X_\beta = 1$$

$$X_{SnA} + X_{SnB} = X_{Sn}$$

$$X_{SnA} X_\alpha = X_{Sn}$$

$$X_{SnA} + X_{SnB} = X_{SnA} X_\alpha$$

$$X_\alpha = 1 + \frac{X_{SnB}}{X_{SnA}}$$

$$X_\alpha = X_\alpha + X_\beta + \frac{X_{SnB}}{X_{SnA}}$$

$$X_\beta = -\frac{X_{SnB}}{X_{SnA}}$$

$$X_{SnB} = C_0 - X_{SnA}$$

$$X_{SnA} = C_0 - X_{SnB} \rightarrow C_0 = X_{SnA} + X_{SnB}$$

$$X_\beta = \frac{X_{SnB} - X_{SnA}}{X_{SnB} - C_0}$$

8

$$X_\alpha = 0.50$$

$$X_E = 0.50$$

$$X_\beta = \frac{40 - 18.3}{97.8 - 18.3} = 0.273$$

$$X_\alpha = 1 - X_\beta = 0.727$$

10

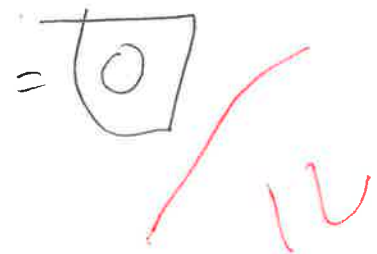
3. You are analyzing a binary Copper-Silver (Cu-Ag) alloy system at a constant atmospheric pressure of 1 atm.

a. Write down the condensed Gibbs Phase Rule equation. (3 points)

b. The alloy is currently exactly at its eutectic temperature and eutectic composition, and it is in the process of solidifying. Calculate the degrees of freedom (F) for the system at this exact moment. (7 Points)

a. $DOF = C - P + 1$

b) $C = 2, P = 3 \rightarrow DOF = 2 - 3 + 1 = 0$



28

Part B: Thermal, Electrical, Magnetic, and Optical Properties

1. Identify the terms defined in the table below from the following list: (8 points)

7

Word bank

heat capacity	temperature coefficient of resistance	ferrimagnetic
specific heat	permeability	ferromagnetic
thermal expansion	magnetic susceptibility	photon
thermal conductivity	magnetic induction	refractive index
electrical resistivity	magnetic moments	total internal reflectance
electrical conductivity	diamagnetic	opaque
dielectric strength	paramagnetic	transparent
thermoelectric effect		translucent

Terms	Definition
Heat Capacity	The amount of heat required to raise the temperature of a material by one degree.
Thermal Expansion	The tendency of a material to expand when heated due to increased atomic vibrations
Ferrimagnetic	Materials with opposing magnetic moments of unequal strength, leading to net magnetization
Magnetic susceptibility	A dimensionless quantity that describes how much a material will become magnetized in response to an external magnetic field
photon	A quantum of light or electromagnetic radiation, carrying energy that depends on its wavelength or frequency.
refractive index	A measure of how much light slows down and bends when entering a material from another medium
translucent	A material that allows light to pass through but scatters it, preventing clear visibility of objects on the other side

electrical resistivity	The maximum electric field a material can withstand without breaking down and becoming conductive
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2. A brass rod is to be used in an application requiring its ends to be held rigid. If the rod is stress-free at room temperature (20°C or 68°F), what is the maximum temperature to which the rod may be heated without exceeding a compressive stress of 172 MPa (25,000 psi)? (6 points)

Assume the following properties for brass:

Modulus of elasticity (E) = 100 GPa (14.6 x 10⁶ psi)

Linear coefficient of thermal expansion (α) = 20.0 x 10⁻⁶ °C⁻¹ (11.1 x 10⁻⁶ °F⁻¹)

$$\sigma = E \epsilon \quad \epsilon = \alpha (T_f - T_i) \rightarrow \sigma = E \alpha (T_f - T_i)$$

$$T_f = T_i + \frac{\sigma}{E \alpha}$$

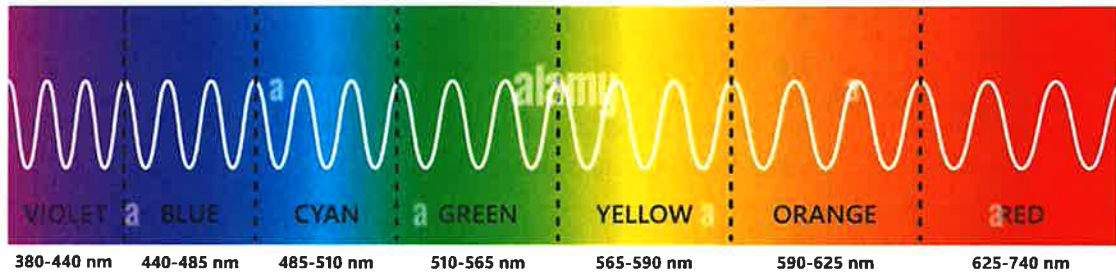
$$\sigma = 0.172 \text{ GPa}$$

$$T_f = 20^\circ\text{C} + 860^\circ\text{C} = 880^\circ\text{C}$$

X 3

3. A photon emitted by a specific light source possesses an energy of 4.42 x 10⁻¹⁹ J.
- Compute the frequency (f) and the wavelength (λ) of this photon in meters. (5 points)
 - Convert your calculated wavelength into nanometers (nm). Using a standard visible light color spectrum chart, determine what color this light appears to the human eye. (3 points)

(Constants: Speed of light, $c = 3 \times 10^8$ m/s; Planck's constant, $h = 6.63 \times 10^{-34}$ Js)



$$i. \quad E = hf \rightarrow f = \frac{E}{h} = \boxed{6.67 \times 10^{14} \text{ s}^{-1}}$$

$$\lambda = \frac{c}{f} = \boxed{4.5 \times 10^{-7} \text{ m}}$$

$$ii. \quad \lambda = 4.5 \times 10^{-7} \text{ m} \times \frac{10^9 \text{ nm}}{\text{m}} = \boxed{450 \text{ nm}}$$

Blue

8

4. The magnetic flux density within a bar of some material is 0.435 tesla at an H field of $3.44 \cdot 10^5 \text{ A/m}$. Compute the following for this material: (a) the magnetic permeability, and (b) the magnetic susceptibility. (c) What type(s) of magnetism would you suggest is displayed by this material? Why? $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ (10 points)

$$\mu = \frac{B}{H}$$

$$B = 0.435 \text{ Tesla}$$

$$H = 3.44 \frac{\text{A}}{\text{m}}$$

$$a) \quad \mu = 1.26 \times 10^{-6} \frac{\text{T} \cdot \text{m}}{\text{A}}$$

$$b) \quad \chi = \frac{\mu}{\mu_0} - 1 = 0.00625$$

c) Paramagnetic

10

Part C: Polymers, Ceramics

21

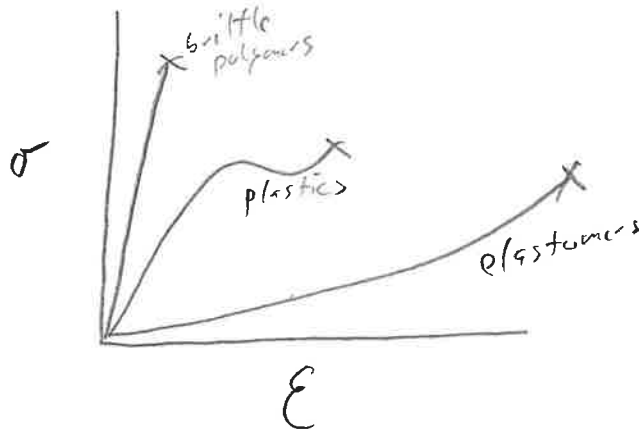
Word Bank

Natural	Atactic
Thermoplastic	Isotactic
Branched	Semi-Synthetic
Synthetic	Crosslinked
Syndiotactic	Amorphous
Crystalline	Thermoset
Semi-Crystalline	Linear

1. Select the word that matches the definition listed below related to polymer classification (4 points).

Polymer Classification	Meaning
<i>Semi-Synthetic</i>	Polymers derived from nature but chemically modified
<i>Thermoset</i>	Polymers that undergo permanent deformation after heating
<i>Branched</i>	A polymer with regular or irregular attachments of side chains to the main polymer's backbone chain
<i>Amorphous</i>	A polymer or polymer structure in which the repeating units have no regular stereochemical configuration.

2. Sketch a stress-strain curve to highlight the behavior of brittle polymers, plastics, and elastomers during tensile testing (8 points)



8

3. You are evaluating the mechanical stability of a brittle ceramic catalyst support for a packed-bed reactor. To determine its flexural strength (σ), you perform a center-point loading test with a support span of $L = 18 \text{ mm}$. The initial tests are performed on **rectangular bar specimens** with width $b = 6 \text{ mm}$ and height $d = 4 \text{ mm}$.

Batch A fractures at an applied force $F = 120 \text{ N}$.

Batch B fractures at an applied force $F = 145 \text{ N}$.

To optimize gas flow, the reactor design team wants to switch to **cylindrical pellets** with radius $R = 3.5 \text{ mm}$, tested over the same 18 mm span.

- Calculate the flexural strength (σ) of each batch from the rectangular specimen tests.
- Using the flexural strength from each batch, determine the maximum applied force (F) the cylindrical pellet can withstand before fracturing. (10 points)

a) A: $\sigma = \frac{3FL}{2bd^2} = 33.75 \text{ MPa}$
 B: $\sigma = \frac{3FL}{2bd^2} = 40.78 \text{ MPa}$

$L = 0.018 \text{ m}$
 $b = 0.006 \text{ m}$
 $d = 0.004 \text{ m}$

b) $\sigma = \frac{FL}{\pi R^3} \rightarrow F = \frac{\sigma \pi R^3}{L}$

$R = 0.0035 \text{ m}$

A: $F = 252.6 \text{ N}$
 B: $F = 305.2 \text{ N}$

10

Part D: Introduction to Macromolecules and Nanoscale Systems

14

1. What is the difference between the top-down and bottom-up methods? (5 points)

Top down methods reduce larger materials into nanomaterials by milling, etching, etc. They are faster but less controlled.

Bottom up methods deposit nanomaterials on a substrate, such as in CVD or ALD. They are slower but more controlled.

5

2. You are developing a wearable biosensor to detect abnormal blood glucose levels in diabetic patients. The sensor's electrical circuit relies on an ultra-thin sheet of graphene (a single atomic layer of carbon). When glucose molecules bind to the graphene surface, they alter its electrical conductivity, triggering an alert.

- a. Classify the single-atom-thick graphene sheet by its nanomaterial dimension (0D, 1D, 2D, or 3D). (3 points)

2D

3

- b. To manufacture this single-layer graphene sheet atom-by-atom to ensure perfect precision, would you choose a "top-down" or "bottom-up" synthesis technique? Why? (5 points)

bottom up.

Top down would not be able to create a sheet with the needed precision. Only bottom up could create a layer of a single atom.

4

c. Glucose is a simple monosaccharide. When the human body has excess glucose, it stores it by linking the molecules together into a large polysaccharide called glycogen. What specific chemical reaction links these glucose molecules together, and what small molecule is released as a byproduct during this process? (5 points)

The chemical reaction is called condensation or dehydration. Water is released as a byproduct,

5

Extra Credit

1. How many known amino groups are there? (3 points)

20

3

2. In the special talk, what deposition method did Dr. Kim use for the MoS₂? (2 points)

Atomic Layer Deposition

2

$$X_A + X_B = C_0$$

$$X_A + X_B = C_0$$

$$X_A + X_B = C_0$$

$$X_A + X_B = C_0$$

$$X_A = X_B$$

$$X_A = X_B$$

$$X_A + X_B = 1$$

$$X_A + X_B = X_A X_B$$

$$X_A = 1 + \frac{X_B}{X_A}$$

$$X_A = X_B + \frac{X_B}{X_A}$$

$$X_B = - \frac{X_B}{X_A} - \left(\frac{C_0 - X_B}{C_0 - X_B} \right)$$

$$X_2 - C_0 = \frac{X_2 - X_1}{X_1}$$